Fiscal Year:	FY 2025	**Task Last Updated:**	FY 08/13/2024
PI Name:	Zupanska, Agata Ph.D.		
Project Title:	From Antarctica to Space: Molecular Response and Physiological Adaptation of Moss to Simulated Deep Space Cosmic Ionizing Radiation and Spaceflight Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
PI Email:	zupanska@live.com	**Fax:**	FY
PI Organization Type:	NON-PROFIT	**Phone:**	352-682-5615
Organization Name:	The SETI Institute		
PI Address 1:	189 Bernardo Ave, 200		
PI Address 2:			
PI Web Page:			
City:	Mountain View	**State:**	CA
Zip Code:	94043-5138	**Congressional District:**	18
Comments:			
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
Contact Monitor:	Zhang, Ye	**Contact Phone:**	321-861-3253
Contact Email:	Ye.Zhang-1@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	As budgeted in the original proposal we hired an OPS Technician, Emily Lockwood. Emily is affiliated with University of Florida and paid from the subaward. Emily works on the project accordingly to proposed effort, 6 months for each year of grant project.		
COI Name (Institution):	McDaniel, Stuart Ph.D. (University of Florida, Gainesville)		
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Task Description:	Rationale: Deep space environments of cosmic ionizing radiation (cIR) from astrophysical events and microgravity (micro-g) are outside of plant evolutionary experience and can impede the vital role of plants in space exploration. Understanding the combined effects of cIR and micro-g on plants at the molecular level is pivotal to engineering deep space-tolerant plants for human space exploration. Most previous research studies each condition alone and uses seeds irradiated with high doses of acute cIR. Hypothesis and Objectives: We propose a Research Investigation (Project Type) into plant growth, development, and physiology (Focus Area) using an International Space Station (ISS)-Flight+Ground Study Component (Platform) to test the hypothesis that combined deep space cIR and micro-g affect plants differently and more profoundly than each environment alone, resulting in distinct molecular responses and physiological adaptation. We aim to detect how the effects of combined simulated chronic deep space cIR and sequential spaceflight micro-g differ from those of cIR or micro-g alone on live plant growth, development, and physiology, as reflected by changes in gene expression, to identify molecular response and physiological adaptation to these environments. Research Approaches and Specific Aims: We will use Antarctic moss Ceratodon purpureus (C. purpureus) which allows us to use irradiated live plants rather than spores in spaceflight. Ceratodon purpureus is tolerant of extreme conditions and is ideal for human spaceflight missions due to its small size, lightweight spores, low-maintenance, and roles in oxygenation, regolith enrichment, and medicine production. We will use a twofold approach: 1) in the ground study protonema will be exposed to simulated Galactic Cosmic Rays (GCRSim, 10 and 100 cGy) and Solar Particle Events (SPESim, 100-1000 cGy) delivered at fractionated dose; 2) in the ISS-Flight study protonema exposed to GCRSim or SPESim on the ground and housed in the NASA Biological Research in Canisters (BRIC)-Light Emitting Diode (LED) will be launched to the ISS and grown in low-Earth orbit (LEO). We will use morphological metrics, RNAseq, and bioinformatics to achieve: AIM1- Discover the dose- and time-dependent effects of simulated deep space cIR on growth, development, and gene expression in C. purpureus. AIM2- Identify the different effects on growth, development, and gene expression in C. purpureus of simulated deep space cIR – singular, or combined with spaceflight micro-g. AIM3- Identify the different effects on growth, development, and gene expression in C. purpureus of spaceflight micro-g – singular, or combined with simulated deep space cIR. Outcome and Significance: This study will: 1) analyze moss growth and development in cIR and micro-g, which will inform plant performance capabilities for human spaceflight exploration; 2) discover moss response and physiological adaptation to cIR and micro-g at the gene expression level, and by extension in other biological systems; 3) discover candidate genes for genetic manipulation to engineer deep-space tolerant plants which will advance the design of Bioregenerative Life Support systems and plant habitats for planetary missions; and 4) extend our results to other multicellular organisms, which will inform health risk management. Relevance to the Plant Biology Element of the NASA Space Biology program: This project addresses: a) the main goal, to “understand plant growth in spaceflight environments and physiological responses to those environments;” b) guiding questions #1 - How does gravity affect plant growth, development and metabolism; #4 - What are the effects of chronic exposure to cosmic radiation on plants; and #5 How do plants sense and react to gravity and what are the molecular mechanisms involved; and c) Plans and Research Solicitations objectives #1 Determine how plants adapt to spaceflight from the molecular to whole plant level using existing hardware (BRIC) on ISS; #3 - Answer basic research questions about plant biology that will enable the future development of a bioregenerative life support system for use by humans in space exploration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In our study we deploy novel approaches in plant space biology research enabling to maximize scientific return. First, unlike other studies relying on model plants, we bring a new organism, moss, which is remarkably adaptive to extreme environments here on Earth and thus is likely to exhibit superior adaptability to deep space environments. Second, in contrast to most other studies which address each deep space environment separately, we will discover the combined effects of both cosmic ionizing radiation and spaceflight microgravity. Third, in contrast with previous research done on irradiated seeds and spores which leads to mutagenesis rather than observation of the effects of deep space cosmic ionizing radiation on plant physiology, our pioneering experimental design sequentially exposes live, actively growing plants to deep space environments. Fourth, we deploy the most accurate simulation of cosmic ionizing radiation of astrophysical origin, exposure to ion beams of compositions and energies best approximating Galactic Cosmic Rays (GCR) and Solar Particle Events (SPE) radiation. Most studies used exposure to electromagnetic wave radiation from UVB, UVC, X-rays, and Gamma-rays delivered at a high-dose in a single pulse (acutely), which poorly simulates deep space cosmic ionizing radiation, which is particulate, chronic, and low-dose at extended fractionations. Upon completion we will deliver moss transcriptomic profiles, growth and development analysis and candidate genes for genetic manipulation to engineer plants more tolerant of cosmic ionizing radiation and spaceflight microgravity. Firstly, this will have a direct impact on advancing plant preparedness for space exploration missions and will help constructing plant varieties that can be incorporated into life support systems and in-space human habitation, enabling long-duration crewed missions beyond Low Earth Orbit. Secondly, since our study will identify the critical plant mechanisms of prevention or repair of the damage caused by ionizing radiation it will potentially result in developing moss-based radiation countermeasures agents of three classes: protectors, mitigators and therapeutics. Such plant compounds could be then used by astronauts during long term interplanetary travel as well as administered to radiation oncology patients or nuclear plant workers accidentally exposed to high radiation doses here on Erath.

Task Progress:

The spaceflight experiment "ANT1 Radiation Tolerance Experiment with Moss in Orbit on the Space Station" (ARTEMOSS) to the International Space Station (ISS) deploys moss Ceratodon purpureus isolate ANT1 collected in Antarctica to address plant growth, development, and physiology, as reflected by gene expression, in the combined environments of deep space ionizing radiation and spaceflight microgravity. ARTEMOSS requires moss tissue fixation in-flight. In Project Year 3, we established deep-freezing as the most effective method of in-flight fixation of moss ANT1 grown on Petri plates, and conducted extensive testing of deep-freeze fixation tailored for the microgravity environment to design ARTEMOSS payload and in-flight operations.
1. Developing a method to fix plant tissue with hydrophobic surfaces. The Science Verification Test (SVT) performed in Project Year 2 showed that initially planned chemical fixation of moss ANT1 tissue in RNAlater was not satisfactory. Through a series of tests, we demonstrated that the waxy, hydrophobic surface of Antarctic moss impeded chemical tissue fixation, and precluded usage of RNAlater coupled with payload hardware Biological Research in Canisters / Petri Dish Fixation Unit (BRIC/PDFU) deployed in standard plant spaceflight experiments. We executed a series of approaches to improve ANT1 fixation, mixing RNAlater with various surfactants or adjuvants, substituting RNAlater with 30, 50 or 70% ethanol, or substituting a chemical fixation with deep freezing. We concluded that deep-freezing the moss tissue on Petri plates provided adequate tissue fixation and allowed for extraction of high-quality RNA suitable for gene expression profiling. Through numerous tests we established a strong preference for use of the much colder -150°C General Laboratory Active Cryogenic ISS Experiment Refrigerator (GLACIER) freezer over Minus Eighty (Degrees Celsius) Laboratory Freezer for ISS (MELFI) to fix a moss payload on the ISS. Our work on Antarctic moss tissue fixation can be translated to other plant species, expanding current techniques of experimentation with plants from extreme terrestrial environments aimed toward advancing human space exploration.

2. Developing a moss payload for the ARTEMOSS spaceflight experiment. We designed in-flight fixation operations for the ARTEMOSS payload that would best secure the transcript profile expressed in orbit while preserving the physical space on the ISS and crew time. Through numerous tests we showed the substantial difference in the moss tissue freezing time resulted from a Petri plate being directly exposed to chilling air or being insulated by other Petri plates in a multi-plate stack. Thus, we arranged the Petri plates in 3-plate stacks to limit variations in the moss tissue freezing time on respective Petri plates. Three (3) biological replicas per treatment occupy each Petri plate position, top, middle, or bottom across a total of five (5) 3-plate stacks to correct for minor discrepancies in the moss tissue freezing time. The 3-plate stacks in ground control are placed on Styrofoam in the -150°C freezer to recreate fixing conditions in microgravity more accurately, where no contact with metal surface is maintained. Thus, we characterized the deep-freeze protocol and in-flight operations for plant payloads that do not require chemical fixation or specialized hardware and that are tailored specifically for the microgravity environment.

3. Passing the EVT and reaching the flight readiness status for ARTEMOSS. The main objectives of the Experiment Verification Test (EVT) were to assess Antarctic moss ANT1 viability, morphology, and level of contamination, as well as the quantity and quality of total RNA extracted from the deep-frozen moss tissue, while best recreating the timeline of the spaceflight ARTEMOSS experiment.

• the EVT execution

1. moss spot-inoculum Petri plate preparation 2. 7-day growth in LIGHT at 23°C 3. 6-day growth in DARK at room temperature/RT (23°C) – transport to NASA Space Radiation Laboratory (NSRL), Brookhaven National Laboratory (BNL) and pre-irradiation ops 4. exposure to Solar Particle Event Simulation/SPESim (at 40cGy and 200cGy) and Galactic Cosmic Ray simulation/GCRSim (at 20cGy and 80cGy) 5. transport in DARK and COLD (4°C) from NSRL, BNL to University of Florida (UF) 6. 3-week storage in DARK and COLD (2.9°C) – recreating a wait for the spaceflight 7. transport of fifteen (15) ARTEMOSS Petri plates to NASA Kennedy Space Center (KSC) 8. onsite payload preparation – generating five (5) 3-plate stacks and the science bag 9. 7-day storage in DARK and COLD (4°C) – recreating pre-flight ops, launch, orbiting, berthing with the ISS 10. 7-day growth in the International Space Station Environmental Simulator (ISSES) running Near Real Time environmental data from the Japanese Experiment Module (JEM) module of the ISS (DARK and ambient) – recreating growth on the ISS 11. the 3-plate stacks transfer to -150°C; the 3-plate stacks laid on the side on Styrofoam- recreating fixation by deep-freeze in GLACIER on the ISS 12. after twenty-four (24) hours, transfer to -80°C for thirty-five (35) days – recreating storage in MELFI on the ISS 13. the EVT payload delivery to the PI 14. the EVT sample analysis – recreating post-flight sample analysis 15. the EVT report delivery

• the EVT outcome

1. PLANT MOSS/VIABILITY – EXCELLENT 100% (39) moss spot-inoculum per treatment, non-irradiated and irradiated, with all 13 spot-inoculum colonies per plate remained viable up to deep-freeze fixation, as reflected in the healthy green color of every colony on every Petri plate throughout, and a substantial elongation of upward filaments during a 7-day growth in the ISSES chamber. 2. PLANT MOSS/MORPHOLOGY – EXCELLENT 100% (39) of moss spot-inoculum colonies per treatment, non-irradiated and irradiated, with all 13 spot-inoculum colonies per plate showed green healthy appearance, no signs of necrotic lesions, wilting, or other morphological aberrations. 3. BACTERIAL OR FUNGAL CONTAMINATION – EXCELLENT No bacterial or fungal contamination on any spot-inoculum, any Petri plate, at any point throughout the EVT. 4. QUANTITY OF TOTAL RNA EXTRACTED – EXCELLENT More than 400ng of total RNA extracted from four (4) spot-inoculums from each Petri plate. The total RNA quantity ranged from 930 to 3690ng as measured on Bioanalyzer, which can be considered a high quantity. 5. QUALITY OF TOTAL RNA EXTRACTED – ACCEPTABLE/UNACCEPTABLE Excellent total RNA quality for 14 samples out of 15 total with RNA Integration Number (RIN) ranging from 7.2-7.9. There was one biological replica among GCRSim 20cGy samples (Rep2) from which extracted RNA showed RIN just below 7.0, RIN=6.9. This RNA showed a very mild RNA degradation and presented no concerns that would warrant the additional round of RNA extractions. This RNA is fully suitable for RNAseq analysis. Two (2) biological replicas of GCRSim 20cGy had RNA with no signs of degradation and RIN of 7.2 and 7.6.

4. ARTEMOSS is ready for launch. ARTEMOSS got authorization to proceed with flight and will be launched on SpaceX-31.

5. ARTEMOSS risk mitigation.

• Since ARTEMOSS is targeting the analysis of radiation-induced damage in spaceflight microgravity, the repair processes on the ground need to be prevented. Keeping moss in cold will halt most metabolic activities and prevent radiation-induced damage from being repaired. Thus, it is essential that irradiated Petri plates designated for the spaceflight are kept between 0-4°C starting from the target room at BNL until the Petri plates are transferred to the

ambient on the ISS. Three (3) different strategies to transport and stow the ARTEMOSS Petri plates in temperatures no higher than 4°C have been tested: 1.) “shipment in cold” with a courier (FedEx); 2.) flying with Petri plates as carry-on luggage aboard the plane and requesting from the Transportation Security Administration (TSA) a hand-check instead of X-ray scanner check; 3.) transporting Petri plates in the two-compartment 12V car refrigerator (BougeRV 12 Volt Car Refrigerator Dual Zone, CRD45(45L) powered by portable power station (Jackery Portable Power Station Explorer 300, 293Wh Backup Lithium Battery) and coupled with thermometer (Govee WiFi Temperature Sensor H5179, Smart Hygrometer Thermometer with App Notification) inserted into the fridge unit next to the Petri plates to monitor the temperature remotely from the iPhone Govee Home app. We concluded that we would drive the car from UF to BNL and back from BNL to KSC and transport ARTEMOSS Petri plates in the car refrigerator using the tested setup.

• Shall SpaceX-31 launch scrub into early 2025, or shall beam time at BNL be unavailable for any reason, we have worked out the access to other forms of particulate radiation:

1.) Proton irradiation at the University of Florida Health Proton Therapy Institute in Jacksonville Florida to substitute for SPESim at NSRL, BNL. The achieved proton energies in the proton radiation therapies range from 70 to 230 MeV and the targeted SPESim ion beam at BNL provides proton beams with descending energies from 150 MeV to 50 MeV. In proton therapies, protons at 150MeV will be used, because protons of this energy were used in other space-relevant studies. In proton therapy, the dose rate will be higher than at BNL with 40cGy dose being delivered under 60s. BNL provides the superior approximation of SPE particulate radiation, but substituting the SPESim ion beam at BNL with the proton beams at the UF Health Proton Therapy Institute will not significantly impact the ARTEMOSS science objectives.

2.) Neutron irradiation at Colorado State University, Fort Collins, Colorado to substitute for GCRSim at NSRL, BNL. The relative biological effectiveness (RBE) factor of 3.0 will be applied to calculate the neutron radiation doses corresponding to the targeted GCRSim doses; irradiation with 60cGy will recreate 20cGy GCRsim dose at BNL and with 180cGy will recreate 80cGy GCRSim dose at BNL. Neutron radiation from energized free neutrons differs from heavy ion radiation. Damage caused by neutrons results from the neutron absorption by atom nuclei and formation of heavier isotopes, some of which are radioactive and could disintegrate. Hydrogen atoms have a relatively high neutron absorption and form deuterium, which can be damaging to cells. Although neutron radiation is space relevant, neutrons are not an accurate replacement for the ion beam used in targeted exposures at BNL. In an unlikely scenario that neutron irradiation will be used, the ARTEMOSS science objectives will be slightly adjusted addressing the effects of the secondary radiation rather than primary collisions with heavy ions. However, ARTEMOSS will still deliver information on the effects of deep space-like irradiation, singular or in combination with the spaceflight microgravity, thus will not deviate much from the original scientific objectives.

Bibliography Type:	Description: (Last Updated: 07/17/2025)
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